

41. Which is mild oxidising agent

- (a) Ag_2O
- (b) KMnO_4
- (c) $\text{K}_2\text{Cr}_2\text{O}_7$
- (d) Cl_2

Answer: A — (a) Ag_2O

Chemical reason: The result is obtained by assigning oxidation numbers before and after the reaction and balancing electron transfer. The species in Ag_2O has the oxidation-state change required by the stated redox conditions, so that choice is correct.

42. The equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ in acidic medium

- (a) 294
- (b) 298
- (c) 49
- (d) 50

Answer: C — (c) 49

Chemical reason: Equivalent weight is calculated as molar mass divided by the number of electrons exchanged per formula unit in the stated redox reaction.

43. The transitional element of III-B group is isomorphous with $\text{Al}_2(\text{SO}_4)_3$. Therefore the compound will be

- (a) Purple
- (b) Blue
- (c) White
- (d) Uncertain

Answer: C — (c) White

Chemical reason: Isomorphous substances crystallize in the same structural type because their ions have similar charge and size. The composition represented by White satisfies that structural relationship.

44. A copper salt is isomorphous with ZnSO_4 , the salt will be

- (a) Paramagnetic
- (b) Diamagnetic
- (c) Ferromagnetic
- (d) None

Answer: A — (a) Paramagnetic

Chemical reason: Isomorphous substances crystallize in the same structural type because their ions have similar charge and size. The composition represented by Paramagnetic satisfies that structural relationship.



45. V_2O_5 is useful as catalyst in

- (a) Manufacture of H_2SO_4
- (b) Manufacture of HNO_3
- (c) Manufacture of Na_2CO_3
- (d) It is not a catalyst

Answer: A — (a) Manufacture of H_2SO_4

Chemical reason: The application follows from the characteristic chemical property of the material—such as catalytic activity, stability, conductivity, or corrosion resistance. That property is specifically associated with Manufacture of H_2SO_4 .

46. $KMnO_4$ in basic medium is reduced to

- (a) K_2MnO_4
- (b) MnO_2
- (c) $Mn(OH)_2$
- (d) Mn^{2+}

Answer: B — (b) MnO_2

Chemical reason: The result is obtained by assigning oxidation numbers before and after the reaction and balancing electron transfer. The species in MnO_2 has the oxidation-state change required by the stated redox conditions, so that choice is correct.

47. When $KMnO_4$ is reduced with oxalic acid in acidic solution, the oxidation number of Mn changes from

- (a) 7 to 4
- (b) 6 to 4
- (c) 7 to 2
- (d) 4 to 2

Answer: C — (c) 7 to 2

Chemical reason: In acidic medium permanganate is reduced from $Mn(+7)$ to Mn^{2+} , so manganese changes from +7 to +2. Oxalic acid is simultaneously oxidized to CO_2 .

48. Nessler's reagent is

- (a) K_2HgI_4
- (b) $K_2HgI_4 + KOH$
- (c) $K_2HgI_2 + KOH$
- (d) $K_2HgI_4 + Hg$

Answer: B — (b) $K_2HgI_4 + KOH$

Chemical reason: Nessler's reagent is an alkaline solution of potassium tetraiodomercurate(II), $K_2[HgI_4]$, used for detecting ammonia. The complex mercury iodide species is the essential reagent component.

49. When ammonium dichromate is heated, the gas formed is

- (a) N_2
- (b) O_2
- (c) H_2
- (d) NH_3

Answer: A — (a) N_2

Chemical reason: On heating ammonium dichromate, the well-known "volcano"



decomposition occurs: $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \rightarrow \text{Cr}_2\text{O}_3 + \text{N}_2 + 4\text{H}_2\text{O}$. Hence nitrogen is the gaseous product.

50. Acidified potassium dichromate on reacting with a sulphite is reduced to

- (a) CrO_2Cl_2
- (b) CrO^{42-}
- (c) Cr^{3+}
- (d) Cr^{2+}

Answer: C — (c) Cr^{3+}

Chemical reason: Acidified dichromate is reduced from Cr(VI) to Cr(III). Therefore the chromium-containing reduction product is Cr^{3+} (commonly represented by a Cr(III) salt in the reaction mixture).

51. The product of oxidation of I^- ion by MnO_4^- in alkaline medium is

- (a) I_2
- (b) IO_3^-
- (c) IO_4^-
- (d) I^{3-}

Answer: B — (b) IO_3^-

Chemical reason: Amphoteric metals/oxides react with both acids and strong bases, often forming soluble hydroxo or oxo complexes in alkali. The species/result IO_3^- is consistent with that behaviour.

52. Identify the statement which is not correct regarding copper sulphate

- (a) It reacts with KI to give iodine
- (b) It reacts with KCl to give Cu_2Cl_2
- (c) It reacts with NaOH and glucose to give Cu_2O
- (d) It give CuO on strong heating in air

Answer: B — (b) It reacts with KCl to give Cu_2Cl_2

Chemical reason: The correct choice is It reacts with KCl to give Cu_2Cl_2 . It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

53. Acidified potassium permanganate solution is decolourised by

- (a) Bleaching powder
- (b) White vitriol
- (c) Mohr's salt
- (d) Microcosmic salt
- (e) Laughing gas

Answer: C — (c) Mohr's salt

Chemical reason: Acidified KMnO_4 is decolourised only by substances that reduce MnO_4^- to Mn^{2+} . Mohr's salt supplies Fe^{2+} , a reducing ion, so it removes the purple permanganate colour.



54. Which of the following oxides is white but becomes yellow on heating

- (a) AgO
- (b) Ag₂O
- (c) FeO
- (d) ZnO

Answer: D — (d) ZnO

Chemical reason: Zinc oxide is white when cold and yellow when hot because heating creates oxygen vacancies/electronic defects; on cooling, it becomes white again.

55. Amalgams are

- (a) Highly coloured alloys
- (b) Always solid
- (c) Alloys which contain mercury as one of the contents
- (d) Alloys which have great resistance to abrasion

Answer: C — (c) Alloys which contain mercury as one of the contents

Chemical reason: An amalgam is an alloy in which mercury is one of the components. The other component may be a metal such as Na, Ag, or Zn.

56. In photography sodium thiosulphate is used as

- (a) Complexing agent
- (b) Oxidising agent
- (c) Reducing agent
- (d) None of these

Answer: A — (a) Complexing agent

Chemical reason: In photographic fixing, thiosulfate ("hypo") forms a soluble complex with Ag⁺, mainly [Ag(S₂O₃)₂]³⁻. This removes unreacted AgBr from the film, so it acts as a complexing agent.

57. The substance that sublimes on heating

- (a) Magnesium chloride
- (b) Silver chloride
- (c) Mercurous chloride
- (d) Sodium chloride

Answer: C — (c) Mercurous chloride

Chemical reason: Mercurous chloride (calomel) is volatile enough to sublime on heating, which distinguishes it from the less volatile salts listed.

58. K₃Fe(CN)₆ is called

- (a) Potassium ferricyanide
- (b) Red prussiate of potash
- (c) Potassium hexacyanoferrate (III)
- (d) All of these

Answer: D — (d) All of these

Chemical reason: K₃[Fe(CN)₆] is potassium ferricyanide, also called red prussiate of potash and potassium hexacyanoferrate(III); all three names describe the same compound.



59. Which of the following will show increase in weight when kept in magnetic field

- (a) TiO_2
- (b) $\text{Fe}_2(\text{SO}_4)_3$
- (c) KMnO_4
- (d) ScCl_3

Answer: B — (b) $\text{Fe}_2(\text{SO}_4)_3$

Chemical reason: The correct choice is $\text{Fe}_2(\text{SO}_4)_3$. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

60. Amongst TiF^{62-} , CoF^{63-} , Cu_2Cl_2 and NiCl^{42-} (Atomic number $\text{Ti}=22$, $\text{Co}=27$, $\text{Cu}=29$, $\text{Ni}=28$). The colourless species are

- (a) CoF^{63-} and NiCl^{42-}
- (b) TiF^{62-} and CoF^{63-}
- (c) Cu_2Cl_2 and NiCl^{42-}
- (d) TiF^{62-} and Cu_2Cl_2

Answer: D — (d) TiF^{62-} and Cu_2Cl_2

Chemical reason: The colour follows from the electronic structure and coordination environment of the transition-metal ion. The species in TiF^{62-} and Cu_2Cl_2 has the characteristic oxidation state/ligand field that produces the observed colour.

